9 Geostationary satellite about Earth has a period of 24 × 60 × 60 = 86400 s.
To find orbital radius,

$$F_G = F_C$$

$$\Rightarrow \frac{GMm_{sat}}{R^2} = m_{sat}R\omega^2$$

$$\Rightarrow \frac{GM}{R^2} = R\left(\frac{2\pi}{T}\right)^2$$

$$\Rightarrow R = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

$$= \sqrt[3]{\frac{(6.67\times10^{-11})(6.0\times10^{24})(86400)^2}{4\pi^2}}$$

$$= 4.23\times10^7\,\text{m}$$

To find gravitational force acting on the astronaut,

$$F = \frac{GMm_{man}}{R^2}$$

$$= \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(60)}{(4.23 \times 10^{7})^2}$$

$$= 13.4 \text{ N}$$